Recap from last lecture

We saw how to calculate fugacity from steam tables or EOS

Fugacity describes the deviation of a real substance from **ideal gas** behaviour

Defined activity coefficient, such that

$$\underline{G} - \underline{G}^{IM} = RT \sum x_i \ln \gamma_i$$

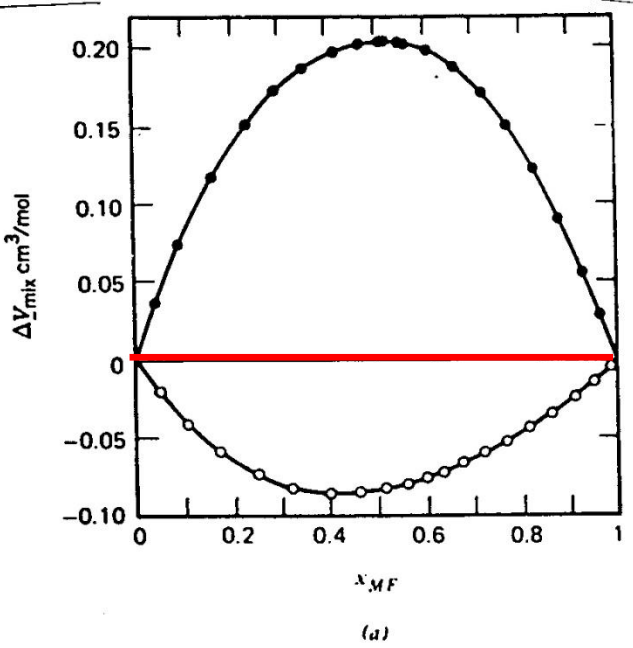
Activity coefficient describes the deviation of a component in a mixture from **ideal mixture** behaviour

$$\Delta \underline{G}_{IM}^{mix} = RT \sum x_i \ln x_i \quad \Delta \underline{S}_{IM}^{mix} = -R \sum x_i \ln x_i \quad \Delta \underline{H}_{IM}^{mix} = \Delta \underline{V}_{IM}^{mix} = 0$$

we want empirical models for γ_i as $\mathrm{fn}(T, x_j)$

- activity coefficient models

[Figures from Sandler]

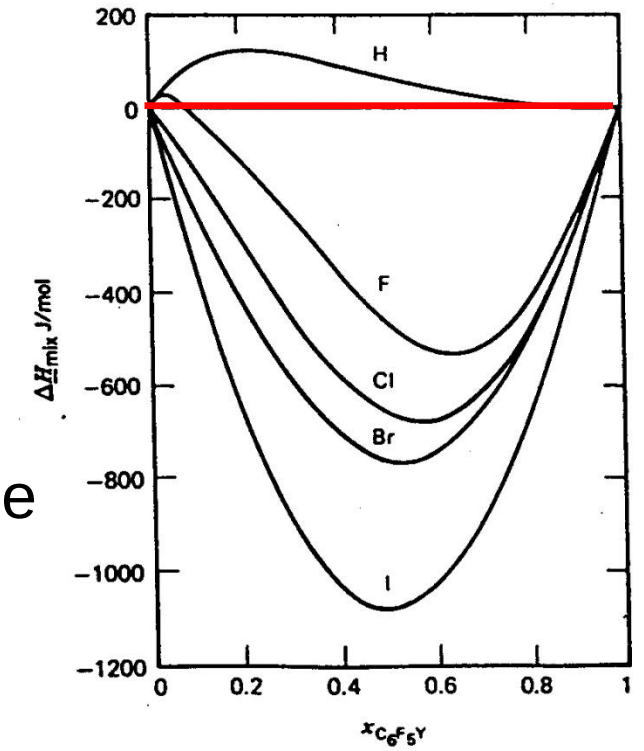


(a) Volume change on mixing at 298.15K; ○ methyl formate + methanol, ● methyl formate + ethanol

IM behaviour

Real mixtures can be far from ideal

Subtle chemical changes make large and unpredicable differences



(b)Enthalpy change on mixing at 298.15K for mixtures of benzene and aromatic fluorocarbons C_6F_5Y, with Y = H, F, Cl, Br and I.

The Activity Coefficient

The activity coefficient describes the **deviation** from **ideal mixture** behaviour:

$$\underline{G} - \underline{G}^{IM} = RT \sum x_i \ln \gamma_i = \underline{G}^{ex}$$

and $$\mu_i - \mu_i^{IM} = RT \ln \gamma_i = \mu_i^{ex}$$

for convenience we call these “excess” properties

N.B. IM $\Rightarrow \mu_i^{ex} = 0 \Rightarrow \gamma_i = 1$

Hence $$\underline{G} = \underline{G}^{IM} + \underline{G}^{ex} = \sum x_i \underline{G}_i + RT \sum x_i \ln x_i + RT \sum x_i \ln \gamma_i$$

and $$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T,P,n_j} = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT \ln x_i + RT \ln \gamma_i$$

Activity Coefficient Models

An activity coefficient model relates the activity coefficient of a species in a mixture to the composition of that mixture (at a given temperature)

We drop the pressure dependence because for “difficult” cases we are dealing with liquids, which have a small molar volume and so little variation of Gibbs energy with pressure.

We are limited by thermodynamics in our choice of activity coefficient model:

Recall the Gibbs-Duhem equation:

For an open system: $$dG = -SdT + VdP + \sum \mu_i dn_i$$

$$\left(\frac{\partial G}{\partial T}\right)_{P,n_i} \quad \left(\frac{\partial G}{\partial P}\right)_{T,n_i} \quad \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j}$$

and we know that $$G = \sum \mu_i n_i$$

differentiating this expression: $$dG = \sum \mu_i dn_i + \sum n_i d\mu_i$$

equating: $$VdP - SdT = \sum n_i d\mu_i$$

or $$\sum n_i d\mu_i = 0$$ at constant T,P

Any activity coefficient model we come up with must obey G-D: this is termed **thermodynamic consistency**.

Now focus on binary systems

At constant T, P (binary mixture): $n_1 d\mu_1 + n_2 d\mu_2 = 0$

$\div RT$ $\quad n_1 d\ln\gamma_1 + n_1 d\ln x_1 + n_2 d\ln\gamma_2 + n_2 d\ln x_2 = 0$

$\div n$ $\quad x_1 d\ln\gamma_1 + x_1 d\ln x_1 + x_2 d\ln\gamma_2 + x_2 d\ln x_2 = 0$

$d\ln x = \frac{dx}{x}$ $\quad x_1 d\ln\gamma_1 + x_1 \frac{dx_1}{x_1} + x_2 d\ln\gamma_2 + x_2 \frac{dx_2}{x_2} = 0$

$dx_1 = -dx_2 \Rightarrow$ $\quad x_1 d\ln\gamma_1 + x_2 d\ln\gamma_2 = 0$

This is just a statement of the Gibbs-Duhem equation in terms of activity coefficients for a binary mixture at constant T, P

Any functions we come up with for μ_1^{ex} and μ_2^{ex} must satisfy this condition

We also require that $\underline{G}^{ex}(x_1 = 0) = \underline{G}^{ex}(x_1 = 1) = 0$

Choose a suitable function for $\underline{G}^{ex}$

Then calculate the μ_is from this – they will obey G-D

The simplest such function is the
"one-constant Margules equation":

$$\underline{G}^{ex} = Ax_1x_2$$

This is where composition dependence appears

A is fn(T) and must be fit to experimental data for $\underline{G}^{ex}$

This is where temperature dependence appears

$\underline{G}^{ex} \to 0$ as $x_1 \to 0$ or $x_2 \to 0$ ($\gamma_i = 1$ for pure i)

$$RT\ln\gamma_1 = \mu_1^{ex} = \left(\frac{\partial G^{ex}}{\partial n_1}\right)_{T,P,n_2} \qquad \underline{G}^{ex} = Ax_1x_2 = \frac{An_1n_2}{(n_1+n_2)^2}$$

$$= \frac{\partial}{\partial n_1}\left(\frac{(n_1+n_2)An_1n_2}{(n_1+n_2)^2}\right)_{T,P,n_2}$$

$$= \frac{An_2}{(n_1+n_2)} - \frac{An_1n_2}{(n_1+n_2)^2} = Ax_2^{\,2}$$

and $\quad RT\ln\gamma_2 = Ax_1^{\,2}$

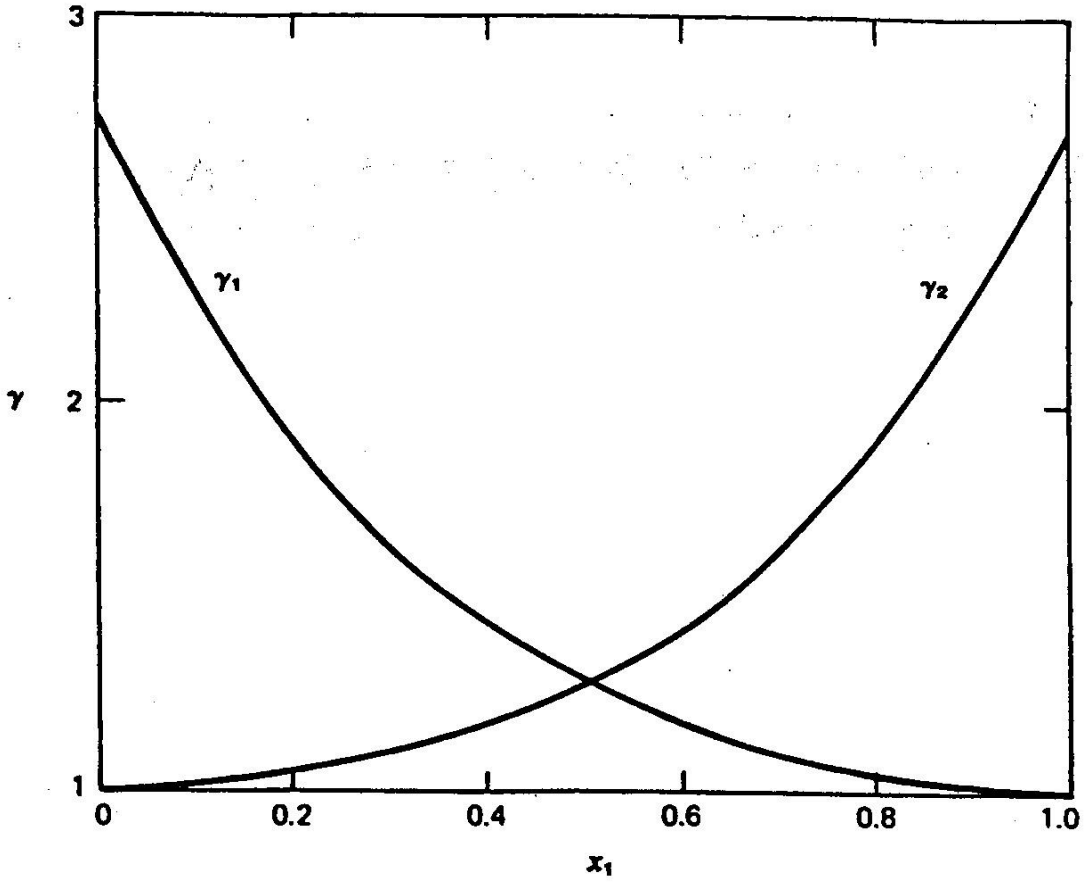
$$d\ln\gamma_1 = \frac{A}{RT}2x_2dx_2 \quad \text{and} \quad d\ln\gamma_2 = \frac{A}{RT}2x_1dx_1$$

$$x_1d\ln\gamma_1 + x_2d\ln\gamma_2 = \frac{A}{RT}(2x_1x_2dx_2 + 2x_2x_1dx_1) = 0$$

The one-constant Margules model is thermodynamically consistent

This does not mean that it will be a “good” activity coefficient model

It is a symmetric model - swapping x_1 and x_2 has no effect on $\underline{G}^{ex}$



[Figure from Sandler]

The activity coefficients for the one-constant Margules equation with (A/RT) = 1

Activity coefficient models need not be symmetric:

a simple asymmetric one is the “two-constant Margules equation”:

$$\underline{G}^{ex} = x_1 x_2 \{A + B(x_1 - x_2)\}$$

This is where composition dependence appears

A and B are both fn(T) and must be fit to experimental data for $\underline{G}^{ex}$

This is where temperature dependence appears

[Student exercise:
show that this model also obeys the Gibbs-Duhem condition]

Physical interpretation of activity coefficient, γ_i

Consider a binary mixture:

$$\gamma_i > 1 \Rightarrow \mu_i > \mu_i^{IM}$$

i.e. the driving force for mass transfer out of the mixture is higher than it would be if the mixture were behaving ideally

This is caused by unlike molecules attracting less strongly than like ones (see later for examples)

It is known as a “positive deviation from ideality”.

$$\gamma_i < 1 \Rightarrow \mu_i < \mu_i^{IM}$$

i.e. the driving force for mass transfer out of the mixture is lower than it would be if the mixture were behaving ideally

Unlike molecules attract more strongly than like ones: "negative deviation from ideality"

$$\lim_{x_i \to 1} \gamma_i = 1$$

We expect big deviations from ideality when a component in a mixture is dilute and small deviation when a component is concentrated

(Think how this is consistent with the Gibbs-Duhem equation)

$$x_1 d \ln \gamma_1 + x_2 d \ln \gamma_2 = 0$$

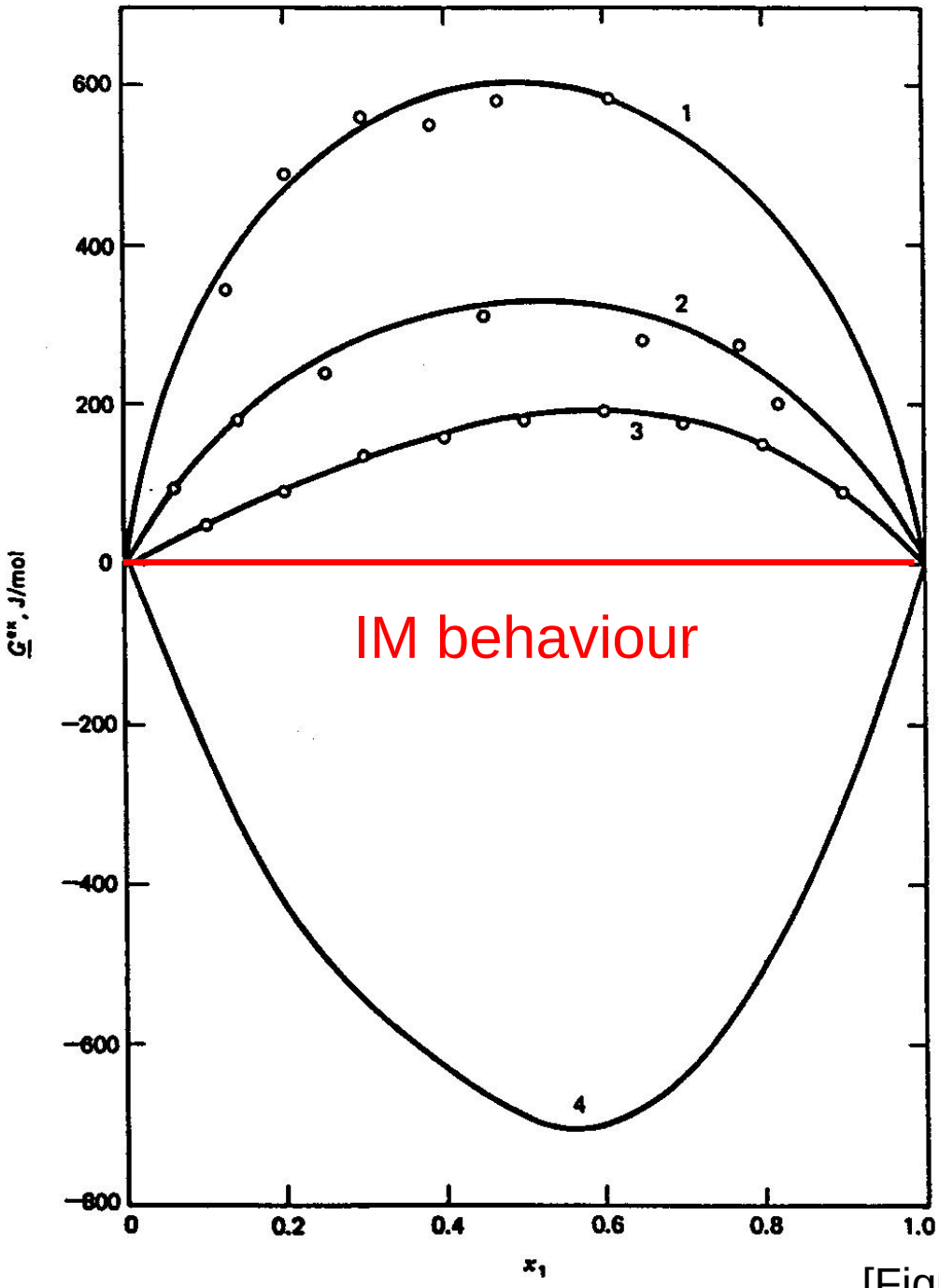
So where do we get these Activity Coefficient Models from?

An activity coefficient model is a function (of composition) for $\underline{G}^{ex}$ at a given T. (Temperature dependence is achieved by the parameter(s) of the model being functions of temperature.)

In favourable cases data is available for $\underline{G}^{ex}$ over a range of compositions and temperatures, and we can fit our model to this data

$\underline{G}^{ex}$ can be calculated, for example, from the vapour pressure exerted by a component in a mixture, from the freezing point depression of a mixture, or from the osmotic pressure exerted by a mixture. It will become clear how this can be done as the course proceeds.

In fact, we are usually calculating fugacities and inferring $\underline{G}^{ex}$



The excess Gibbs free energy for:

Curve 1: Trimethyl methane (1) and benzene at 0ºC.

Curve 2: Trimethyl methane (1) and carbon tetrachloride at 0ºC.

Curve 3: Methane (1) and propane at 100K.

Curve 4: Water (1) and hydrogen peroxide at 75ºC.

[Figure from Sandler]

Unisim has a library of A, B parameters as fn(T) for a range of mixtures.

The pressure dependence of $\underline{G}^{ex}$ is usually small for liquid mixtures (why?) and so it is normal to ignore it

Alternatively you can input your own parameters into Unisim

Note: just as for EOSs for mixtures you only need data for each binary pair in a multicomponent mixture.

Parameters from the binary mixture are then combined to give an Activity Coefficient Model for the complete mixture. We will see how this is done